

WiFi-CSI sensing for human presence and machine activity in industrial environments

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Abstract—The fields of human presence detection (HPD) and people counting (PC) from WiFi-Channel State Information (CSI) represent constantly evolving areas of research within the scientific community. This study aims to expand the research frontiers by exploring the application of HPD and PC in an industrial environment characterized by its considerable dimensions, the presence of obstacles and machines, and harsh propagation channel conditions. Using a convolutional neural network (CNN) incorporating CSI measurements, we demonstrated the ability to distinguish between human presence and industrial machine activity with the same model, achieving an accuracy of over 99%. In addition, a study was conducted to count up to three people, even when machines were on, with an accuracy of over 92%. Another experiment was conducted with the objective of assessing not only the feasibility of PC in a naturally challenging environment but also the impact of the WiFi channel bandwidth in this particular application. The results of this study highlight the potential of WiFi-CSI sensing to address a significant challenge in human activity detection and estimation: its application in industrial environments.

Index Terms—WiFi sensing, channel state information, human presence detection, people counting, deep learning, industrial environments.

I. INTRODUCTION

In the current context, where Internet of Things (IoT) technologies drive the creation of intelligent environments, the use of WiFi-Channel State Information (CSI) signals for human presence detection (HPD) and people counting (PC) is presented as a non-intrusive and ubiquitous solution. The CSI allows HPD and PC to take advantage of the variations in the properties of the wireless communication channel caused by the presence and movement of people, achieving high accuracy compared to traditional metrics such as Received Signal Strength Indicator (RSSI) [1].

Concerning its application, wireless signals offer advantages over other most common techniques, such as image processing or wearable devices. Some key advantages that WiFi-CSI-based data offers over image processing include enhanced user privacy, detection through walls, and reliable usability in low-light conditions. User-free interaction is a key advantage compared to HPD systems based on wearable devices. Moreover, WiFi-CSI-based systems offer a cost-effective solution, requiring minimal infrastructure and no compromise to privacy.

WiFi-CSI demonstrates considerable promise in applications related to HPD and PC. Initial research has centered on small environments, such as laboratories, smart homes, and offices, with the objective of enhancing automation, healthcare, and security [2], [3]. However, the technology's scalability and compatibility with existing network infrastructures suggest significant potential for application in industrial environments, where its benefits could be substantial. In environments such as factories and warehouses, this technology has the potential to facilitate real-time monitoring of employee presence, optimize personnel management, and ensure compliance with safety protocols. For instance, it could detect unauthorized access to restricted areas or overcrowded work areas, triggering alerts in the event of incidents. Furthermore, in production plants, its capacity to analyze human flow could assist in identifying bottlenecks in processes, enabling dynamic adjustments in personnel distribution and shift planning to enhance operational efficiency. Another significant application under study is its use in emergency management, where it would provide real-time data on human presence and the number of people within a structure, facilitating evacuations and reducing risks.

Nevertheless, industrial environments present specific challenges for HPD and PC applications due to their characteristics. These spaces are usually large, with irregular distribution of furniture, machinery, electrical devices, operators, and different surfaces that generate dispersion, reflection, and multipath propagation of the signals, hindering the system's accuracy. In addition, interference caused by other wireless devices, coupled with the conducted and radiated electrical noise characteristic of these areas, further complicates the accurate detection of signal changes caused by people. Therefore, a study on the impact of the industrial environment and the use of machinery on the human presence detection capabilities of deep learning (DL) models is considered essential.

To address these challenges, performing human measurements in real industrial environments in different situations is crucial. In addition, it is key to study the impact of WiFi signal parameters in the DL models, such as the channel bandwidth or the number of subcarriers, to optimize the use of available resources in each context. Integrating advanced signal processing and deep learning techniques is paramount to enhancing the adaptability and efficiency of these systems

in industrial environments.

This paper studies HPD and PC applications in industrial environments using WiFi-CSI data by means of a convolutional neural network (CNN). The main contributions of the paper are:

- Experiments involving measurements on industrial environments.
- Distinguishing between human presence and machine activity in industrial environments using a CNN for CSI data.
- Study of the influence of machines activity on people counting. We analyze if there is a difference in the precision of the counting of people moving freely in an industrial environment when the machines are on and when they are off.
- Finally, the impact of channel bandwidth on the accuracy of PC models is analyzed. Differences in the accuracy of counting up to three people moving in an industrial environment are evaluated considering bandwidths of 20 and 80 MHz.

The article is organized into eight sections, the first being the Introduction. In Section II (Related Works), previous studies on HPD and PC based on CSI, as well as works on predicting or classifying activities related to industrial environments, are reviewed. Section III (Channel State Information) discusses the nature of CSI. Section IV (Data Collection) details the methodology employed to collect data in a challenging industrial environment. Section V (Description of the Experiments) describes the experiments performed to realize the HPD and PC in the industrial environment. Section VI (Data Preprocessing and CNN Model) details the data preprocessing steps and the architecture of the CNN-based model. Section VII (Results and Discussion) presents and discusses the classification results of the different experiments based on the classification metrics used. Finally, Section VIII (Conclusions) summarizes the study's achievements.

II. RELATED WORK

WiFi-CSI-based systems have undergone significant advancements, with HPD [4]–[6] and PC [7]–[12] being two of the most prominent applications. Various techniques have been developed to address noise, interference, and environmental variability. These include the acquisition, preprocessing, and denoising of CSI data [13], [14], as well as signal transformations and feature extraction [12]. Subsequently, DL models are employed to identify specific signal patterns and accurately associate them with human activities or other relevant classifications [15]. In our approach, the preprocessing stage incorporates channel phase processing to enhance the accuracy of the DL model [16].

These developments have enabled the creation of highly accurate HPD and PC studies. However, significant challenges remain regarding scalability in large spaces and calibration in complex environments. Most of the studies conducted have focused on small to medium-sized spaces, such as classrooms, laboratories, and rooms, which limits the applicability of the

models in a wider variety of scenarios [10], [11], [16]–[21]. In the field of machine sensing and industrial environments, research efforts have primarily centered on fault identification and prediction, with particular attention given to applications such as solar panels [22], [23] and electric motors [24], [25]. This identification process utilizes various signals and parameters to train DL models. Another focus is on using WiFi-CSI signals in conjunction with DL tools for detecting household appliances using the impulsive noise generated by these devices [26]. However, to the best of our knowledge, the study of HPD and PC in industrial environments has not been addressed, which represents an opportunity for further research in this field.

III. CHANNEL STATE INFORMATION

CSI is a metric that enables the assessment of the status of a communication link. It represents the Channel Frequency Response (CFR) of the channel in Orthogonal Frequency Division Multiplexing (OFDM) used in many communication systems. The CSI is used in communications to estimate the signal degradation in the transmission channel. This allows the receivers to adapt to the channel variations and improve the reception. The CSI is expressed as a function of the CFR of K subcarriers, as outlined below:

$$\text{CSI} = [H_1, \dots, H_K], \quad (1)$$

where H_k is the CSI value for the k -th subcarrier, defined as:

$$H_k = |H_k|e^{j\varphi_k}, \quad (2)$$

where $|H_k|$ represents the amplitude, and φ_k denotes the phase.

If X_k and Y_k refer to the transmitted and received signals for the k -th subcarrier, respectively, their relationship is expressed as:

$$Y_k = H_k X_k + N, \quad (3)$$

where N is the complex additive white Gaussian noise in the communication channel. The CSI, as depicted in Equation 1, is estimated at the receiver. Nevertheless, this estimation is vulnerable to errors stemming from channel impairments and synchronization issues. The phase, notably, exhibits a high degree of sensitivity to these errors. Consequently, when employing the phase for detection purposes, it is essential to apply phase sanitization techniques to minimize these effects [16], [27], [28]. The estimated phase $\hat{\varphi}_k$ for the k -th subcarrier at the receiver can be expressed as:

$$\hat{\varphi}_k = \varphi_k + 2\pi \frac{m_k}{P} \Delta t + \beta + Z, \quad (4)$$

where:

- Δt is the time offset at the receiver,
- m_k is the index of the k -th subcarrier,
- β accounts for the random phase offset introduced by the Carrier Frequency Offset (CFO),
- Z represents measurement noise, and
- P is the size of the Fast Fourier Transform (FFT).

Accurate phase estimation is primarily hindered by three factors:

- 1) $\frac{2\pi m_k}{P} \Delta t$, caused by Sampling Frequency Offset (SFO) and Symbol Timing Offset (STO),
- 2) β , due to the CFO, and
- 3) Z , caused by the measurement noise.

To enhance detection performance using phase information, the Time Smoothing and Frequency Rebuild (TSFR) method [16] has been employed in this study. This method removes phase shifts through a multi-stage preprocessing approach. It begins with linear regression, followed by time-domain filtering using the Savitzky-Golay filter. Lastly, phase correction based on thresholds is applied to resolve subcarrier discontinuities.

IV. DATA COLLECTION

The measurements used in this research were made in the Laboratory of Mechanical Design and Advanced Manufacturing (LADIMMA) of the Engineering Institute of the National Autonomous University of Mexico (UNAM). This laboratory (Figures 1 and 2) is of large dimensions (25 × 25 m) and includes industrial machines such as lathes and milling machines, among others.



Fig. 1. LADIMMA industrial environment with the presence of three people marked inside yellow ovals and location of transmitter Raspberry inside the white oval.

The CSI measurement acquisition system was implemented using four Raspberry Pi 4 devices, an access point (AP), and two laptops. One of the Raspberry Pi was set as a file transfer protocol (FTP) server as the transmitter. One laptop was connected to the AP and acted as an FTP client, sending requests to download 4K video from the FTP server to generate network traffic in the WiFi channel. Meanwhile, the other three Raspberry Pi were used as receivers to monitor the RF channel. The Nexmon CSI tool was used to capture the CSI data and was configured on all the Raspberry Pi. Finally, the second laptop, connected via Ethernet to the three receivers, was used to manage the CSI data acquisition through a MATLAB program.

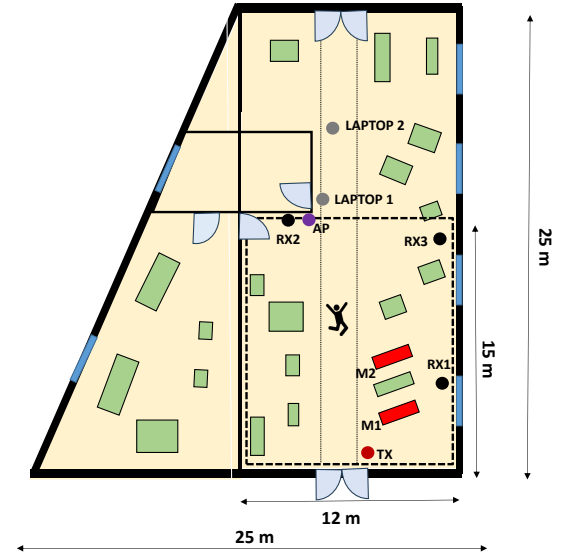


Fig. 2. Schematic of the LADIMMA industrial environment. The transmitter is located at the red dot, the black dots represent the receivers, the purple dot indicates the AP, and the gray dots show the laptop's position. The people move within the dotted dashed zone. The machines M1 and M2 are industrial lathes.

Each recorded CSI measurement lasted 60 seconds. Channel 13 in the 2.4 GHz band and the 802.11n standard were used for 20MHz bandwidth measurements, and channel 40 in the 5 GHz band with the IEEE 802.11ac standard was used for 80MHz bandwidth measurements.

Although LADIMMA is large, the participants were instructed to move freely within the area bounded by the dashed line in Figure 2, which is approximately 12 × 15 meters (dimensions that are still large). The measurements utilized in this study were collected on five non-consecutive days. On the first day, data were collected for both HPD and PC experiments. The second day was devoted exclusively to HPD, while the remaining three days focused on PC. Consequently, two days of measurements are available for HPD, forming the HCSI-1 and HCSI-2 sets. In contrast, the PC experiments are distributed over four measurement days, resulting in sets designated as PCSI-1, PCSI-2, PCSI-3, and PCSI-4. Notably, the PCSI-4 set corresponds to the day on which PC tests were conducted using two machines (M1 and M2 of Figure 2). Table I shows the names of the sets created, the days on which the measurements were taken, the bandwidths used, and the applications intended for their development

TABLE I
DISTRIBUTION OF THE DATA IN DIFFERENT SETS, BANDWIDTHS USED, AND APPLICATIONS.

Day	Set Name	BW (MHz)	Application
Day 1	HCSI-1	80	HPD
	PCSI-1	20 and 80	PC
Day 2	HCSI-2	80	HPD
Day 3	PCSI-2	80	PC
Day 4	PCSI-3	20 and 80	PC
Day 5	PCSI-4	20 and 80	PC

The measurements were organized into the following classes:

- Empty (E): Area with no people and all machines turned off.
- 1, 2, 3 people (1_P, 2_P, 3_P): One, two, or three individuals moving freely within the defined area
- Machine working alone (M_W_A): Measurement of an industrial lathe (M1) working without people in the area.
- Machine working with an operator (M_W_Men): Measurement of an industrial lathe in work, with an operator nearby.

Each measurement obtained with a bandwidth of 20 MHz contains 64 subcarriers and a range of between 15K and 70K CSI frames, while the 80 MHz measurements contain 256 subcarriers and between 20K and 100K CSIs. Each receiver recorded a different amount of CSI values in each measurement, which is attributed to a low signal-to-noise ratio due to its position within the environment and the spectral occupancy of the Wi-Fi channel.

V. DESCRIPTION OF THE EXPERIMENTS

The objective of the experiments conducted in this work is to evaluate the possibility of performing HPD and PC in industrial environments. To do this, the impact of factors such as the presence of running machines, the industrial environment itself, and WiFi signal bandwidth are tested. Consequently, three experiments have been designed:

- Experiment 1: Determine whether it is possible to differentiate between human presence and machine activity in an industrial environment. Four different classes were defined: E, 1_P, M_W_A, and M_W_Men (Section IV). These measurements have a bandwidth of 80 MHz.
- Experiment 2: Evaluate the feasibility of performing PC of up to three people in an industrial environment, both with and without machine activity. To this end, measurements are performed in two scenarios: (1) machines off, where from zero to three people are counted, and (2) machines on, where from one to three people are counted. In each case, people move freely within the measurement area. The measurements employed in this study have a bandwidth of 80 MHz.
- Experiment 3: Compare the impact of bandwidth on the accuracy of people counting. To this end, classification models are trained using measurements made with a bandwidth of 20 MHz. The data set includes three days in two different situations: 1) with machines off, recording counts from zero to three people and one to three people, and 2) with machines on, with counts from one to three people. A comparison of the PC accuracy between 20 and 80 MHz bandwidths is facilitated by the correspondence in terms of days between these measurements and those of Experiment 2.

VI. DATA PREPROCESS AND CNN MODEL

According to the IEEE 802.11 standards, OFDM modulation assigns a specific number of null, pilot, and data

subcarriers, depending on the bandwidth. During the preprocessing stage, the null subcarriers were eliminated, leaving 56 useful subcarriers (data + pilots) at 20 MHz and 208 at 80 MHz. Note that, in the case of 80 MHz, the number of null subcarriers was 48, as the duplicate non-high throughput (non-HT) transmission mode [29] was used, operating with 4 channels of 20 MHz, each with its corresponding null subcarriers.

The resulting CSI values were organized in two-dimensional matrices and structured according to time and frequency. The columns represent the frequency information given by each subcarrier, while the rows contain different CSI values, which provide the time information.

In this study, a CNN was implemented as a DL algorithm to obtain classification models in HPD and PC. This CNN (Figure 3) is a sequential model designed to classify activity data. It starts with three convolutional layers, using 32, 64, and 128 filters with 5x5, 3x3, and 1x1 kernel sizes, respectively. Each convolutional layer is followed by the Mish activation function and batch normalization to stabilize and accelerate training. A dropout layer with a rate of 0.2 is included to reduce overfitting. The feature maps are then flattened and passed through two fully connected dense layers with 256 and 128 units, both utilizing the Mish activation function and batch normalization. Finally, the output layer applies a softmax activation function to classify the data into the number of classes defined in each case. The model is compiled with a Sparse Categorical Cross-entropy loss function and optimized using Adam. Training and validation are performed using a stratified k-fold cross-validation approach to ensure balanced class representation across folds and provide robust evaluation. The model performance is assessed using accuracy as the primary metric, alongside a confusion matrix for deeper analysis.

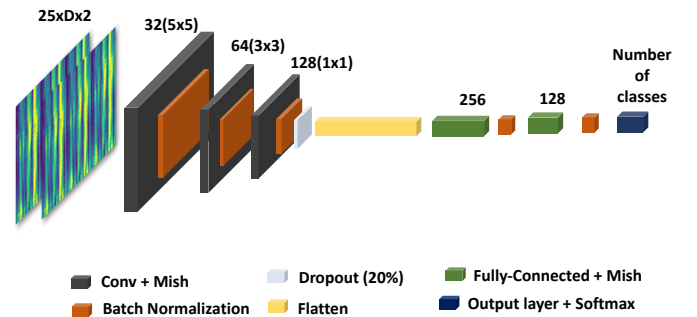


Fig. 3. CNN Model.

The input data to the CNN comprises both amplitude and phase information from the CSIs, which have been previously preprocessed with the TSFR method. These data are structured as two-channel images or matrices with dimensions ($S \times D \times 2$), where S corresponds to the number of CSIs per matrix, D denotes the number of useful subcarriers, and 2 represents the amplitude and phase channels. This representation employs the capabilities of the CNN to process spatial and temporal

information in CSI analysis. The data in each experiment is divided into 60% for training, 20% for validation, and 20% for testing.

In this study, the accuracy and F1 Score metrics are utilized to evaluate the performance of the classifier. Additionally, several of the generated confusion matrices will be analyzed. The confusion matrix provides a comprehensive analysis of a classifier's performance by showing the relationship between the model's predictions and the actual values. Through it, it is possible to identify correct and incorrect classifications in each category, enabling the calculation of key metrics such as accuracy, precision, specificity, recall, and F1 Score. Accuracy, which calculates the proportion of correct predictions over the total number of instances, provides a general measure of the model's performance. However, due to its potential bias in imbalanced datasets, it is complemented by the F1 Score, a metric that offers a more balanced evaluation, particularly useful in scenarios with class imbalance. The F1 Score combines precision (the model's ability to avoid misclassifying negative and positive instances) and recall (its ability to correctly identify all positive instances) through the harmonic mean [30] [31]. This integration allows for a more robust and comprehensive assessment of the model, ensuring a precise analysis of its effectiveness, even in contexts with uneven class distributions.

VII. RESULTS AND DISCUSSION

In this section, we analyze the results of the experiments described above. The models are trained independently with data from each set, which includes the information collected by the three receivers. In all the experiments, the number of CSI per array was set to $S=25$, and the number of D subcarriers was specified accordingly in each case.

A. Experiment 1

This experiment evaluates the possibility of distinguishing between human and machine activity in an industrial environment. It uses measurements made during two days: HCSI-1 and HCSI-2. The used bandwidth is 80 MHz, leading to a total of 208 subcarriers for matrices with dimensions (25, 208, 2). The classification system employed includes four distinct classes: E, 1_P, M_W_A, and M_W_Men. Table II provides a comprehensive overview of the number of matrices assigned to each class, along with the accuracy (Acc.) levels and F1 Score achieved for each day of the measurement period.

TABLE II
EXPERIMENT 1. NUMBER OF MATRICES PER CLASS, ACCURACY, AND F1 SCORE.

Set Name	Classes				Metrics	
	E	1_P	M_W_A	M_W_Men	Acc. (%)	F1 Score
HCSI-1	15045	13094	13067	12949	99.95	0.999
HCSI-2	15347	14263	9489	12989	99.97	0.999

The results demonstrate high accuracy on both days, effectively differentiating between human activity, machine activity,

and the combination of both activities. In all cases, the model demonstrates an accuracy level that exceeds 99%, along with an F1 Score that remains above 0.99. These metrics are derived from the test set, suggesting a high degree of reliability and the model's capacity to generalize effectively to unseen data.

B. Experiment 2

The matrices utilized in this experiment correspond to four days of measurements, with a bandwidth of 80 MHz and 208 subcarriers. The size of each matrix is therefore (25, 208, 2). From PCSI-1 to PCSI-3, the environment has no machine activity. However, on PCSI-4, two machines are on at all times (industrial lathes, M1 and M2, as shown in Figure 2). The main difference between a machine on and a machine working is the state of the motor. In the first case, the motor is not engaged; in the second case, it is engaged and can then manufacture the workpieces fed into it. The operator usually turns on the machine during large periods but only engages it when he is manufacturing a piece.

The PC was up to 3, with four classes involved on the first two days: E, 1_P, 2_P, and 3_P. On PCSI-3, there were no measures of E, with classes ranging from one to three people. On PCSI-4, there were also no E measures, with classes being 1_P, 2_P, and 3_P. Table III presents the number of matrices by class, along with the accuracy and F1 Score achieved in the classification. "MA" means machine activity.

TABLE III
EXPERIMENT 2. NUMBER OF MATRICES PER CLASS, ACCURACY, AND F1 SCORE.

Set Name	MA	Classes				Metrics	
		E	1_P	2_P	3_P	Acc. (%)	F1 Score
PCSI-1	No	22210	27284	34819	28235	95.72	0.957
PCSI-2	No	20518	13346	16673	15014	99.83	0.998
PCSI-3	No	-	23857	50812	43227	92.53	0.925
PCSI-4	Yes	-	13101	15385	13486	98.86	0.989

The results show accuracy values above 92% in all cases, which is considered a solid classification performance. Figure 4 shows the confusion matrix for PCSI-3, selected for analysis because it shows the lowest accuracy among the cases evaluated. On this day, there are only three classes, and it is observed that the biggest errors of the model are made when it tries to classify situations with two and three people. The 11% of the data corresponding to 2_P confuses them with 3_P, and vice versa, 3.9% of the 3_P matrices classify them as 2_P. In the case of PCSI-4, despite the presence of two machines on, the model manages to perform PC with 98.86% accuracy. The errors are analyzed in Experiment 3. The F1 Score values obtained serve to confirm the effectiveness and equilibrium of the model.

The results indicate that slightly higher accuracy is achieved in the presence of running machines compared to the scenarios where the machines are off, which is the opposite of what might be expected. Although this is an area that requires further study, the results are promising as they suggest that

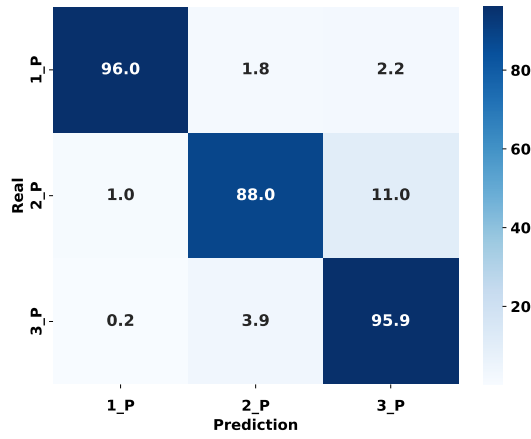


Fig. 4. Confusion matrix of Experiment 2 for PCSI-3 80 MHz.

the influence of running machines does not invalidate the use of CSI-WiFi to accurately perform the PC process. This was an initial concern at the start of this study, given the lack of this type of research in the literature.

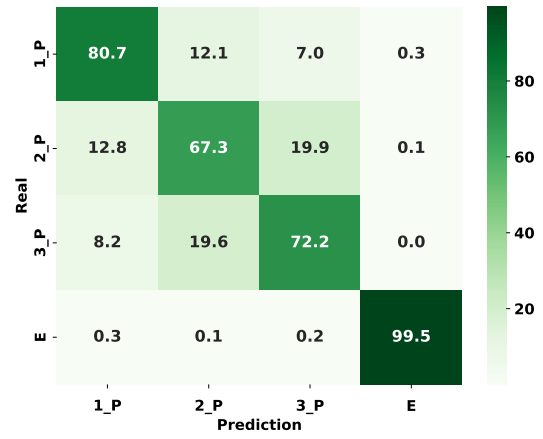
C. Experiment 3

In order to analyze the impact of bandwidth on PC accuracy, this experiment compares 20 and 80 MHz measurements, or in other words, compares 56 and 208 subcarrier matrices. In Experiment 2, the counting accuracy of up to three people was obtained for 80 MHz. In this experiment, 20 MHz measurement models from PCSI-1 to PCSI-3 (machines off) and PCSI-4 (machines on) are trained and compared with the results obtained above. The size of the matrices for 20 MHz is (25, 56, 2), and for 80 MHz is (25, 208, 2). The classes for PCSI-1 are four: E, 1_P, 2_P, and 3_P. On PCSI-3, the analysis is limited to three classes: 1_P, 2_P, and 3_P were recorded since no measurements were taken for the E class. On PCSI-4, similar to PCSI-3, but with machines on, the classes are: 1_P, 2_P, and 3_P. The difference between PCSI-3 and PCSI-4 is the machine activity. While in PCSI-3, the machines are off; in PCSI-4, two machines are on. Table IV shows the accuracy achieved for 20 MHz, and the 80 MHz (these last ones are the same as Experiment 2) result from the previous experiment is added.

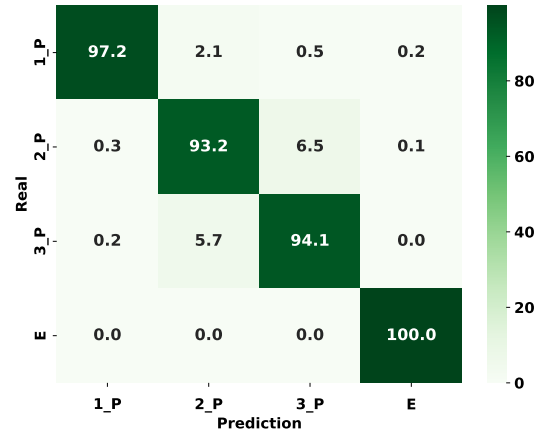
TABLE IV
EXPERIMENT 3. ACCURACY FOR 20 AND 80 MHz.

Set Name	Classes	20 MHz(%)	80 MHz(%)
PCSI-1	E, 1_P, 2_P, 3_P	79.89	95.72
PCSI-3	1_P, 2_P, 3_P	82.89	92.53
PCSI-4	1_P, 2_P, 3_P	83.54	98.86

The results indicate that the accuracy on PCSI-1 for 20 MHz is 79.89%, on PCSI-3 it reaches 82.89%, and on PCSI-4 it is 83.54%. In all cases, these values are lower than the results achieved with the 80 MHz measurements. This difference in performance was expected since the 80 MHz matrices contain four times more frequency information than



(a) 20 MHz



(b) 80 MHz

Fig. 5. Confusion matrices of Experiment 3 with PCSI-1 for 20 and 80 MHz.

the 20 MHz matrices, providing a higher density of frequency data and contributing to improved model performance. Despite the difference, the accuracy results for 20 MHz are considered acceptable. Figure 5 shows the PCSI-1 confusion matrices for both bandwidths to analyze the main classification errors.

In the 20 MHz case, the classes that show more errors in the classification are two and three people. The 19.9% of the class 2_P is classified as 3_P, and 19.6% of the 3_P data is classified as 2_P. For 80 MHz, although the errors are less significant, they are similar to the previous case. The 6.5% of the 2_P data are classified as 3_P, and 5.7% of the 3_P data are classified as 2_P. For both bandwidths, the best detecting class is E.

After reviewing the confusion matrices for PCSI-4 (Figure 6), it is clear that the 20 MHz case presents significant challenges in the classification of 2_P, with only 66.6 % of the data correctly classified. A significant part, 27.2%, is misclassified as 3_P and 6.1% as 1_P. The 80 MHz case has already been analyzed in a previous experiment, and although it shows high accuracy values, for 3_P, it only classifies 81.5% of the data correctly, and 18.5% of the data is confused with 2_P.

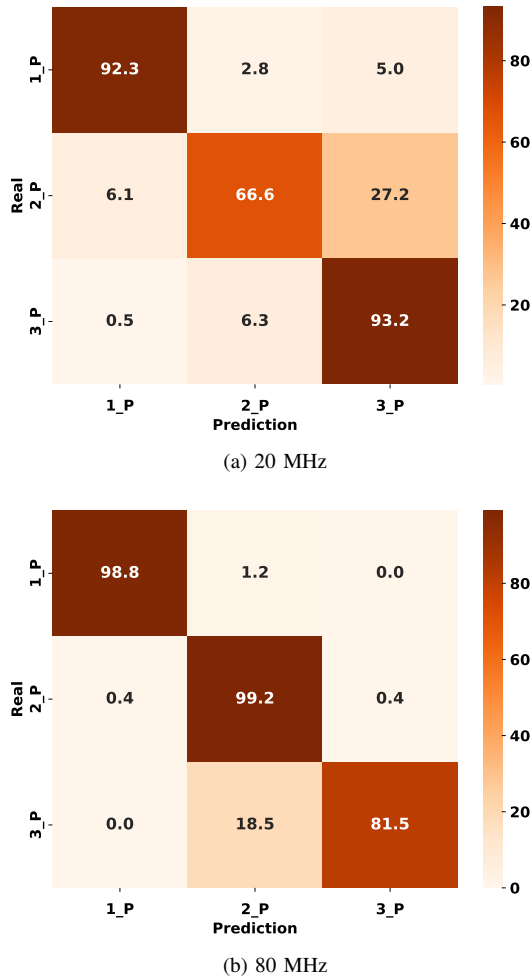


Fig. 6. Confusion matrices of Experiment 3 with PCSI-4 for 20 and 80 MHz. People moving freely within the defined area with two industrial lathes on.

The latest WiFi versions, including WiFi 7 [32], offer enhanced bandwidth capabilities of up to 320 MHz, thereby optimizing the performance of wireless networks. However, this augmentation in bandwidth necessitates an increase in processing demands, making it crucial to assess its impact. Specifically, experiments that examine the impact of bandwidth on PC are imperative to evaluate how it affects the accuracy and reliability of these systems.

VIII. CONCLUSIONS

This research analyzes the HPD and PC of up to three people in an industrial environment, both with machines on and off, using WiFi-CSI signals. To this end, a measurement system was implemented in an industrial environment to obtain CSI, and a CNN was employed to train the classification models. The study attained an accuracy level of over 99% in the HPD-related experiment, successfully detecting the presence of a walking person and precisely classifying an operator engaged in machine-related activities. This validates the capacity to discern between machine operation and human presence. In the context of PC, the accuracy surpassed 92.53%

when the machines were off and attained 98.86% when they were on. This suggests that the presence of machines does not have a negative impact on people counting. However, future research should confirm this fact, as well as testing with a more significant number of people to verify where the system loses accuracy. In terms of bandwidth, the results with 80 MHz were found to be superior, although those with 20 MHz were also deemed acceptable. This study represents a starting point for future research involving human presence detection, machine activity recognition, and people counting. Its application has great industry potential, contributing to improving productivity and safety in future industrial environments.

ACKNOWLEDGMENT

The authors extend their sincere gratitude to the Basque Government (grant IT1436-22) and the Spanish Government for their financial support through the THERESA project (grant PID2021-124706OB-I00, funded by MICIU/AEI/10.13039/501100011033 and co-financed by the ERDF, "A way of making Europe"). They also acknowledge CONAHCYT for the support provided through student grant No. 928123, as well as UNAM for its long-term activity support grants and the UNAM-PAPIIT program (grant IN102025).

Special thanks to the UNAM Engineering Institute, particularly the LADIMMA laboratory team, for their invaluable assistance in conducting measurement campaigns within industrial environments.

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